AI-Based Water Quality Monitoring and Pool Maintenance System

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Project Proposal Report

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DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

Swimming pools are widely used in hotels, sports complexes, and recreational facilities, making proper water quality management essential for ensuring swimmer safety and hygiene. However, many swimming pool facilities still rely on fixed maintenance schedules and manual monitoring practices that do not consider variations in pool usage. High swimmer density can accelerate water contamination, increase turbidity, and reduce the effectiveness of disinfecting chemicals. As a result, traditional monitoring approaches often fail to respond efficiently to real-time usage conditions, which can lead to poor water quality and potential health risks for swimmers.

Despite the availability of sensor-based monitoring systems, most existing solutions focus only on measuring water parameters and generating threshold-based alerts. These systems rarely consider the influence of swimmer density and pool usage patterns on water quality degradation. Therefore, there is a clear research gap in developing intelligent systems that integrate real-time usage monitoring with predictive maintenance planning.

This research proposes a vision-based crowd-aware maintenance scheduling and optimization module as part of a smart Water Quality Monitoring and Pool Maintenance System. The system uses CCTV cameras to capture real-time video streams of the swimming pool environment and applies computer vision techniques to detect and count swimmers. A deep learning–based object detection model is used to estimate swimmer density, which is then used to calculate a Crowd Density Level (CDL). The estimated crowd density is correlated with water quality parameters such as pH, turbidity, temperature, and chlorine levels to predict water quality degradation and generate dynamic maintenance schedules and chemical dosing recommendations.

The proposed system utilizes technologies such as computer vision, deep learning models, OpenCV, and Python-based data analysis to perform real-time monitoring and intelligent maintenance planning. The expected outcome is an automated system capable of optimizing pool maintenance operations while reducing unnecessary chemical usage.

Socially, the system improves swimmer safety by ensuring better water quality management, while commercially it provides a scalable solution for hotels, sports complexes, and recreational facilities to enhance operational efficiency and sustainable pool management.

Keywords

Computer Vision, Crowd Density Estimation, Smart Pool Management, Predictive Maintenance, CCTV-Based Monitoring

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LIST OF ABBREVIATIONS

Abbreviation	Meaning
AI	Artificial Intelligence
CCTV	Closed-Circuit Television
CDL	Crowd Density Level
CV	Computer Vision
IoT	Internet of Things
ML	Machine Learning
OpenCV	Open-Source Computer Vision Library
pH	Potential of Hydrogen
TDS	Total Dissolved Solids
WQMS	Water Quality Monitoring System

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INTRODUCTION

Swimming pools are widely used in hotels, sports complexes, recreational facilities, and educational institutions as spaces for leisure, exercise, and social activities. Maintaining proper water quality and ensuring safe swimming conditions are essential for protecting public health. However, many swimming pool facilities still rely on traditional monitoring methods and fixed maintenance schedules that do not consider real-time pool usage patterns. As the number of swimmers increases, contaminants such as body oils, sweat, cosmetics, and microorganisms enter the water, causing rapid deterioration of water quality. This can lead to increased turbidity, reduced chlorine effectiveness, and the spread of water-borne infections.

The problem becomes more critical in facilities with high visitor turnover such as hotels and public swimming pools. During peak hours, many swimmers can occupy the pool simultaneously, accelerating the degradation of water quality. However, existing maintenance systems typically follow predetermined schedules for cleaning and chemical dosing without considering how many swimmers used the pool. As a result, maintenance activities may occur too late or too early. Late maintenance can lead to unsafe water conditions, while unnecessary maintenance wastes chemicals, energy, and operational resources.

Several technological solutions have been introduced to improve swimming pool monitoring. Modern systems use water quality sensors to measure parameters such as pH levels, turbidity, temperature, and chlorine concentration. These systems can generate alerts when water parameters exceed safe thresholds. Although sensor-based monitoring provides valuable data, it does not consider the influence of swimmer density on water quality degradation. Most systems react only after water quality becomes unsafe rather than predicting when maintenance should occur.

To address these limitations, this research proposes a vision-based crowd-aware maintenance scheduling and optimization module as part of a smart Water Quality Monitoring and Pool Maintenance System. The proposed approach utilizes CCTV cameras installed around the swimming pool to monitor swimmer activity in real time. Computer vision techniques are used to detect and count swimmers from video streams, allowing the system to estimate crowd density. This information is combined with water quality data to predict when maintenance actions such as filtration cleaning and chemical dosing are required. By integrating swimmer density analysis with maintenance planning, the proposed system enables dynamic maintenance scheduling that adapts to actual pool usage conditions.

The proposed research aligns with Sustainable Development Goal (SDG) 3: Good Health and Well-Being, which aims to promote healthy lives and ensure safe recreational environments. By improving water quality management and reducing contamination risks, the system contributes to healthier public facilities. Furthermore, the research aligns with the Centre of Excellence in Artificial Intelligence (CEAI) research cluster by applying artificial intelligence and computer vision techniques to improve smart facility management systems.

1.1 Background and Context

Swimming pools are widely used in hotels, sports complexes, schools, and recreational centers as spaces for relaxation, exercise, and social interaction. Maintaining proper water quality in these facilities is essential for protecting public health and ensuring a safe swimming environment. However, many swimming pool management systems still rely on fixed maintenance schedules and manual observation to monitor pool conditions. These traditional methods often fail to reflect the actual usage of the pool, especially during peak hours when many swimmers use the facility simultaneously.

High swimmer density significantly affects water quality because contaminants such as sweat, body oils, cosmetics, and microorganisms are introduced into the water. This contamination can lead to increased turbidity, reduced chlorine effectiveness, and the spread of waterborne infections. According to global health studies, poorly maintained swimming pools can cause skin infections, eye irritation, and gastrointestinal diseases. Therefore, maintaining proper water quality is critical for ensuring swimmer safety and public health.

Existing pool monitoring systems mainly rely on water quality sensors that measure parameters such as pH level, turbidity, chlorine concentration, and temperature. While these systems provide useful data, they usually generate alerts only when water parameters exceed safe thresholds. This reactive approach does not consider swimmer density or pool usage patterns, which are key factors influencing water quality deterioration.

To address these limitations, this research proposes a vision-based crowd-aware maintenance scheduling and optimization module as part of a smart Water Quality Monitoring and Pool Maintenance System. The proposed system uses CCTV cameras installed around the swimming pool to capture real-time video streams and detect swimmers using computer vision techniques. By estimating the number of swimmers in the pool, the system can determine crowd density and correlate it with water quality degradation patterns. This information is then used to generate dynamic maintenance schedules and optimize chemical dosing, enabling a more efficient and proactive maintenance strategy.

This research contributes to Sustainable Development Goal (SDG) 3: Good Health and Well-Being by improving public safety and hygiene in recreational water facilities. Furthermore, the study aligns with the Centre of Excellence in Artificial Intelligence (CEAI) research cluster by applying artificial intelligence and computer vision technologies to develop intelligent monitoring and decision-support systems for smart facility management.

1.2 Literature Review

Recent developments in artificial intelligence and computer vision have enabled automated monitoring systems to analyze real-time visual data for safety and environmental management applications. Object detection is a key technique in computer vision that allows systems to identify and locate objects within images or video streams. Deep learning algorithms such as YOLO (You Only Look Once) have demonstrated strong performance in real-time object detection tasks, including crowd monitoring, surveillance, and human activity analysis.

Several studies have applied computer vision techniques to detect and count people in crowded environments such as transportation hubs, shopping malls, and public events. These systems use deep learning models to estimate crowd density and analyze human movement patterns. Research shows that YOLO-based models are particularly effective for real-time detection because they provide high accuracy and fast processing speeds, making them suitable for surveillance and monitoring systems.

In parallel, many studies have explored the use of sensor-based monitoring systems to track water quality parameters in swimming pools and other water bodies. These systems commonly use IoT sensors to measure factors such as pH levels, turbidity, temperature, and chlorine concentration. Sensor-based systems can automatically notify operators when water quality levels exceed predefined safety thresholds. However, these systems typically focus only on chemical measurements and do not analyze the influence of human activity on water quality degradation.

Although both crowd monitoring and water quality monitoring technologies have been studied extensively, there is limited research that integrates these two approaches. Most existing swimming pool monitoring systems operate independently from crowd analysis systems. As a result, maintenance decisions are usually made without considering swimmer density or pool usage patterns. This limitation highlights the need for an intelligent monitoring solution that combines computer vision-based crowd detection with water quality management to support predictive maintenance strategies.

1.1 Research Gap

Despite the availability of water quality monitoring technologies and computer vision-based crowd detection systems, there is a lack of integrated solutions that combine these technologies for intelligent swimming pool management. Most existing pool monitoring systems rely on chemical sensors and threshold-based alerts, which only indicate problems after water quality has already deteriorated.

Another major limitation is that current maintenance practices are largely reactive rather than predictive. Maintenance activities such as cleaning and chemical dosing are typically performed according to fixed schedules, regardless of how many swimmers have used the pool. This approach may lead to unsafe swimming conditions during periods of high usage or unnecessary maintenance during low usage periods.

Furthermore, many existing monitoring systems do not utilize real-time visual data to estimate swimmer density and analyze its impact on water quality. Without understanding the relationship between crowd density and contamination levels, pool operators cannot accurately predict when maintenance should occur.

Therefore, there is a clear research gap in developing a system that integrates computer vision-based swimmer detection with water quality monitoring to enable crowd-aware maintenance scheduling. This research aims to address this gap by proposing an intelligent system that uses

CCTV cameras to estimate swimmer density and correlate it with water quality parameters to generate optimized maintenance recommendations. By combining visual monitoring with predictive analysis, the proposed solution can improve maintenance efficiency, enhance swimmer safety, and reduce unnecessary resource usage in swimming pool facilities.

2. OBJECTIVE

2.1 Main Objective

The main objective of this research is to develop a CCTV-based crowd-aware maintenance scheduling module that uses computer vision techniques to detect and count swimmers in real time and optimize swimming pool maintenance activities based on crowd density and water quality conditions.

2.2 Specific Objectives

- To design a computer vision-based system that detects and counts swimmers using CCTV camera footage in the swimming pool environment.
- To estimate real-time crowd density levels by analyzing the number of swimmers detected in the pool area.
- To analyze the relationship between swimmer density and water quality degradation, including parameters such as turbidity, chlorine levels, and ph.
- To develop a dynamic maintenance scheduling mechanism that recommends cleaning and chemical dosing based on crowd density and water quality data.
- To evaluate the effectiveness of the proposed system in improving maintenance efficiency, reducing chemical overuse, and ensuring safe swimming conditions.

3. METHODOLOGY

This section describes the research approach, technical methods, tools, and evaluation techniques used to develop the proposed CCTV-based crowd-aware maintenance scheduling module for swimming pool management.

Research Approach

This research follows a Design Science Research (DSR) approach, which focuses on designing and evaluating an innovative technological solution to solve a real-world problem. The problem addressed in this study is the lack of intelligent monitoring systems that consider swimmer density when planning swimming pool maintenance activities.

Using this approach, research will design a computer vision-based system that detects swimmers using CCTV cameras, estimates crowd density, and generates optimized maintenance schedules based on real-time pool usage patterns. The proposed system will then be evaluated through experimental testing using collected data to measure its effectiveness in improving maintenance decision-making.

Technical Methods and Algorithms

The proposed system will use computer vision and machine learning techniques to analyze real-time video streams captured by CCTV cameras installed around the swimming pool.

First, the video stream will be processed using a deep learning–based object detection model, such as the YOLO (You Only Look Once) algorithm, which is widely used for real-time human detection tasks. The model will identify and count swimmers present in the pool area.

Once swimmers are detected, the system will calculate a Crowd Density Level (CDL) based on the number of swimmers detected within the monitored area. This density information will be combined with water quality parameters such as pH levels, turbidity, temperature, and chlorine concentration collected from water quality monitoring sensors.

Using this combined data, a crowd-aware maintenance scheduling algorithm will analyze the relationship between swimmer density and water quality degradation trends. Based on this analysis, the system will generate dynamic recommendations for maintenance activities such as pool cleaning, filtration cycles, and chemical dosing.

Tools and Platforms with Justification

Several software tools and platforms will be used to implement the proposed system.

The system will be developed using Python, which provides powerful libraries for computer vision and machine learning. The OpenCV library will be used for image and video processing

tasks such as frame extraction and object tracking. The YOLO Deep Learning model will be used for swimmer detection because of its high accuracy and real-time processing capability.

A MySQL database will be used to store swimmer detection results, water quality parameters, and system logs. MySQL was chosen because it provides reliable data management and supports efficient data querying for analysis.

For the system interface, a web-based dashboard will be developed using modern web technologies. This dashboard will allow pool managers to monitor swimmer density, water quality parameters, and maintenance recommendations in real time.

Data Collection and Ethical Considerations

Data for this research will be collected using CCTV cameras installed in the swimming pool environment. The video data will be used only for detecting swimmer presence and estimating crowd density. Personal identification of individuals will not be performed, ensuring that privacy and ethical guidelines are maintained.

Additionally, water quality data such as pH level, turbidity, temperature, and chlorine concentration will be collected from water quality monitoring sensors installed in the swimming pool.

All collected data will be used solely for academic research purposes, and appropriate data protection measures will be implemented to ensure that the privacy of individuals using the swimming pool is respected.

Validation Strategy and Performance Metrics

The proposed system will be evaluated using experimental testing to measure its performance and effectiveness. Several performance metrics will be used to assess the accuracy and efficiency of the system.

The swimmer detection model will be evaluated using accuracy, precision, recall, and detection rate to measure how effectively the system identifies swimmers from CCTV footage. The crowd density estimation will also be compared with manually observed swimmer counts to verify its reliability.

Additionally, the effectiveness of the maintenance scheduling module will be evaluated by analyzing how well the system predicts maintenance requirements based on swimmer density and water quality data.

The results of these evaluations will help determine whether the proposed system can improve swimming pool maintenance efficiency, reduce unnecessary chemical usage, and enhance overall swimmer safety.

4. HIGH-LEVEL SYSTEM ARCHITECTURE

Overview of the Proposed System

The proposed Water Quality Monitoring and Pool Maintenance System integrates multiple intelligent modules to monitor swimming pool conditions, ensure swimmer safety, and optimize maintenance operations. The system combines computer vision, machine learning, and sensor-based monitoring to provide real-time insights into pool usage, water quality conditions, and safety risks.

The system is composed of four major modules: a crowd-aware maintenance scheduling module, a water quality prediction module, a drowning detection module, and a garbage intrusion detection module. These modules work together to collect environmental data, analyze pool activity, and generate automated alerts and maintenance recommendations.

The overall system architecture follows a modular design where each component performs a specific task while sharing data through a centralized processing layer and database. This architecture ensures flexibility, scalability, and efficient system integration.

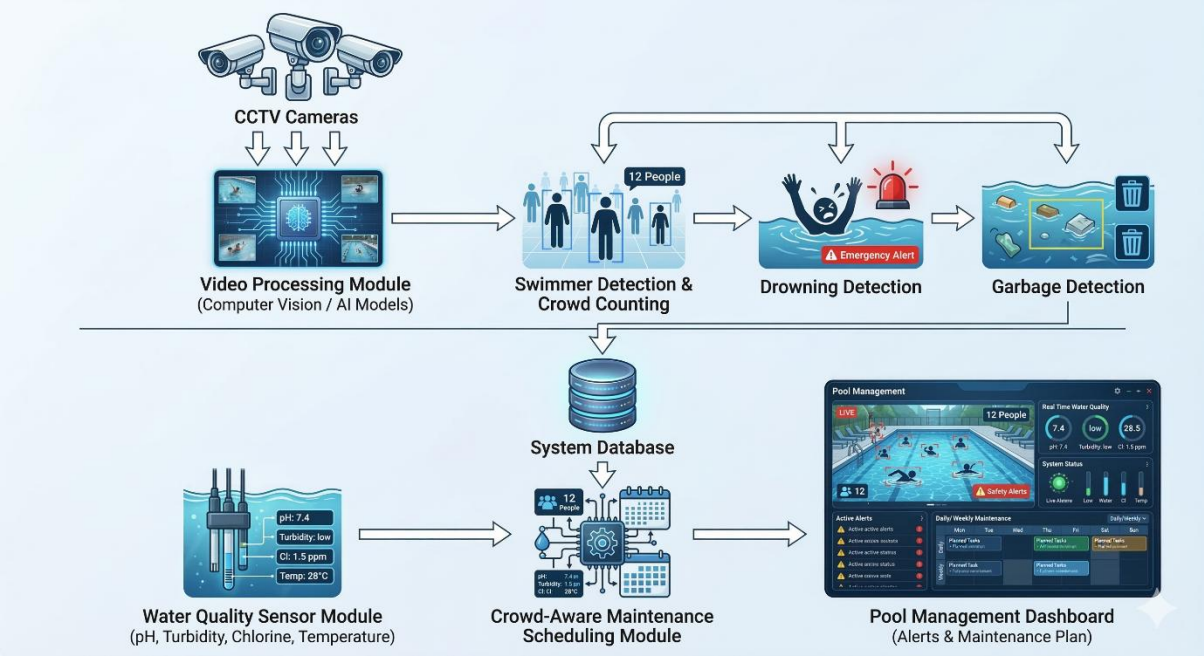


Figure 4.1 High-Level System Architecture of the Proposed Water Quality Monitoring and Pool Maintenance System

System Components

The proposed system consists of the following four main components:

1. Crowd-Aware Maintenance Scheduling Module *(MY Component)*

This module analyzes swimmer density and water quality conditions to optimize swimming pool maintenance operations. Using swimmer count data obtained from the vision system and water quality data collected from sensors, the module determines how pool usage affects water quality degradation.

Based on this analysis, the module generates dynamic maintenance schedules, including chemical dosing recommendations, filtration cycles, and cleaning operations. By replacing traditional fixed maintenance schedules with usage-aware planning, the system improves operational efficiency and reduces unnecessary chemical consumption.

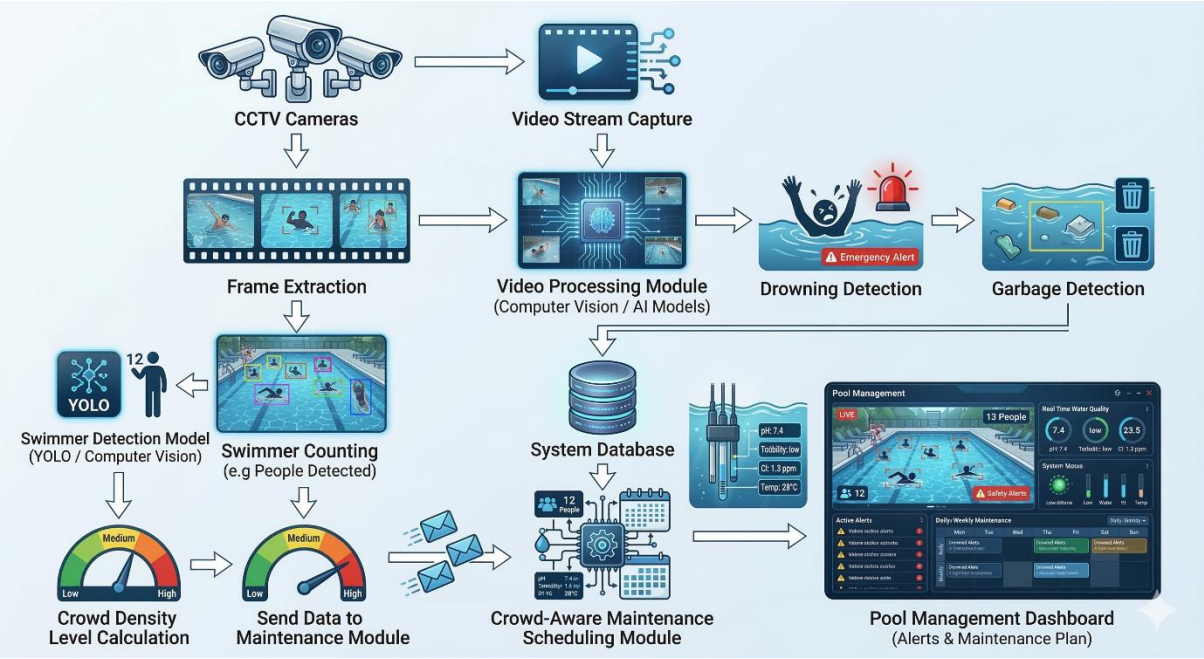


Figure 4.2 CCTV-Based Swimmer Detection and Crowd Counting Process

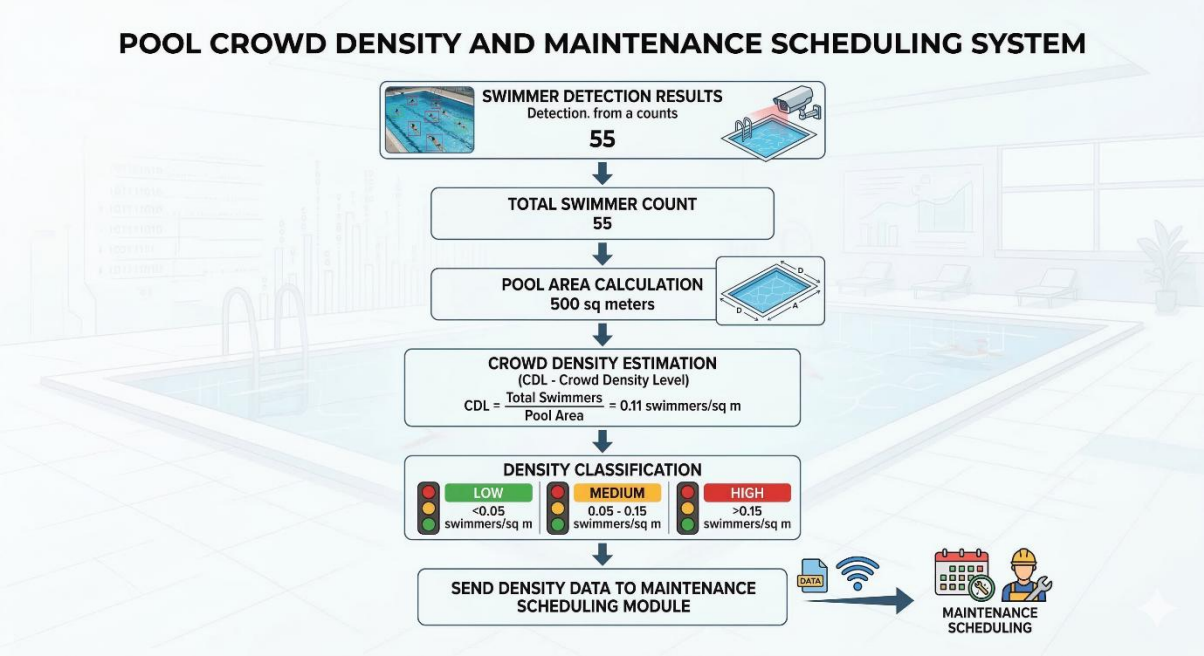


Figure 4.3 Crowd Density Estimation Workflow

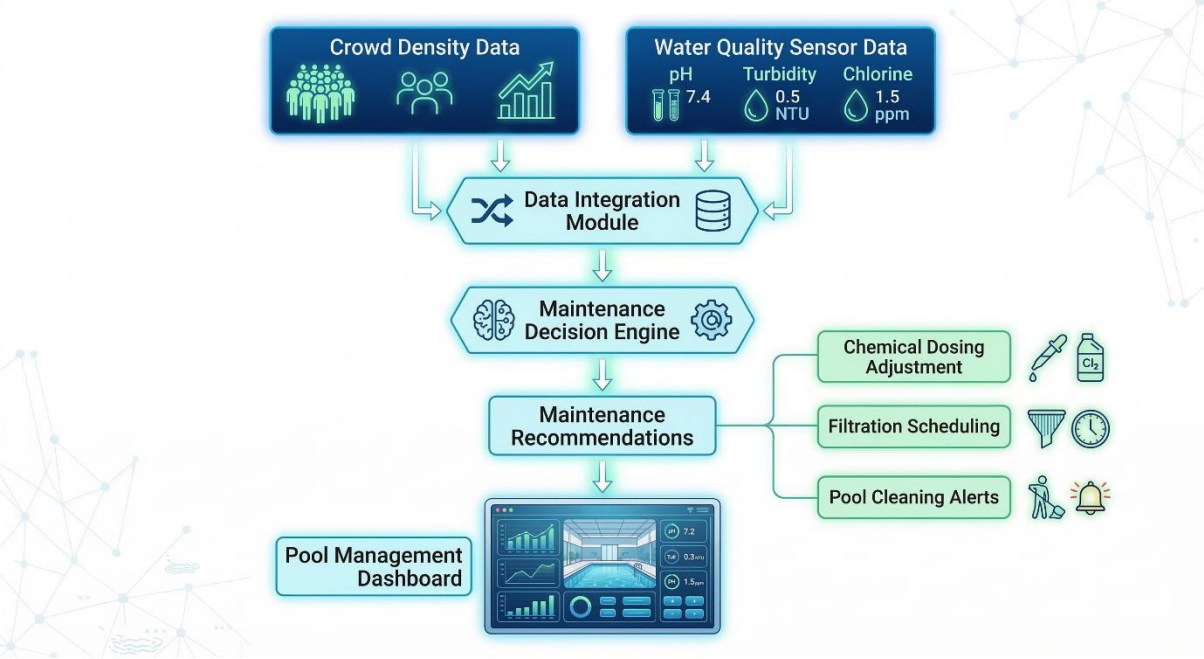


Figure 4.4 Crowd-Aware Maintenance Scheduling Process

2. AI-Based Water Quality Classification and Prediction Module

This module analyzes data collected from water quality sensors, including parameters such as pH level, turbidity, temperature, total dissolved solids (TDS), and chlorine concentration.

Machine learning models are used to classify water quality conditions into **Safe, Warning, and Critical** categories.

The system also evaluates the importance of each water parameter in predicting water quality conditions, enabling better early warning systems and improving decision-making for pool maintenance.

3. Swimmer Behavior and Drowning Detection Module

This component uses computer vision techniques to monitor swimmer activities in real time. Video streams from CCTV cameras are analyzed using deep learning models to identify abnormal swimmer behaviors such as prolonged inactivity, distress movements, or sinking patterns.

When potentially dangerous behavior is detected, the system automatically generates emergency alerts to notify lifeguards or pool management, enabling faster response and improved swimmer safety.

4. Garbage Intrusion Detection Module

This module monitors the swimming pool environment to detect garbage objects entering the pool area. A computer vision-based object detection model identifies items such as plastic bottles, wrappers, and other waste materials.

Additionally, the system analyzes human actions to determine whether the garbage intrusion was intentional or accidental. If high-risk events are detected, alerts are sent to pool management to enable rapid cleaning actions and maintain water hygiene.

Component Interaction

All system modules interact through a centralized data processing platform and database. CCTV cameras provide video streams that are processed by both the drowning detection module and the garbage detection module, while swimmer detection results are shared with the crowd-aware maintenance scheduling module.

At the same time, water quality sensors continuously collect environmental data that is processed by the water quality prediction module. The maintenance scheduling module combines swimmer density information with water quality predictions to determine optimal maintenance schedules.

All generated alerts, predictions, and recommendations are displayed on a centralized pool management dashboard, allowing operators to monitor pool conditions and respond to issues in real time.

Data Flow Explanation

The system data flow begins with **data acquisition from CCTV cameras and water quality sensors**. Video streams are analyzed using computer vision algorithms to detect swimmers,

identify abnormal behaviors, and detect garbage objects. Sensor data is collected and processed to determine water quality status.

These data streams are then integrated within the system's processing layer, where machine learning models analyze patterns and generate predictions. The crowd-aware maintenance module uses these results to determine maintenance needs.

Finally, all system outputs are presented through the management dashboard, which provides real-time monitoring, alerts, and maintenance recommendations.

Scalability and Security Considerations

The proposed architecture is designed to support scalability, allowing additional cameras and sensors to be integrated into the system as needed. This enables deployment in larger swimming pool facilities or multi-pool environments.

From a security perspective, video data is processed only for activity detection and swimmer counting without identifying individuals. Access to system data and the monitoring dashboard is restricted to authorized personnel to ensure data privacy and operational security.

5. USER REQUIREMENTS

5.1 Stakeholder Identification

The proposed crowd-aware pool maintenance system involves several stakeholders who interact with the system or benefit from its outputs.

Stakeholder	Role in the System
Pool Managers	Monitor pool conditions and receive maintenance recommendations through the dashboard.
Maintenance Staff	Perform cleaning, chemical dosing, and filtration tasks based on system alerts.

Lifeguards	Ensure swimmer safety and monitor crowd conditions using the system insights.
Swimming Pool Users	Benefit from improved water quality and safer swimming conditions.
System Administrators	Maintain the system infrastructure, database, and software components.

These stakeholders ensure the effective operation and maintenance of the intelligent pool monitoring system.

5.2 Functional Requirements

Functional requirements describe the core features and capabilities that the system must perform.

1. The system shall capture real-time video streams using CCTV cameras installed around the swimming pool.
2. The system shall detect and count swimmers in the pool using computer vision–based object detection models.
3. The system shall calculate the Crowd Density Level (CDL) based on the number of swimmers and the pool area.
4. The system shall collect water quality data from sensors including pH, turbidity, chlorine level, and temperature.
5. The system shall integrate crowd density information with water quality sensor data.
6. The system should analyze the combined data to determine the appropriate pool maintenance requirements.
7. The system shall generate dynamic maintenance recommendations including:
    - Chemical dosing adjustments
    - Filtration scheduling
    - Pool cleaning alerts
8. The system shall display real-time pool monitoring data and maintenance recommendations through a pool management dashboard.

9. The system shall store swimmer count, sensor readings, and maintenance records in a centralized database.

10. The system shall notify pool management when crowd density or water quality exceeds predefined thresholds.

5.3 Non-Functional Requirements

Non-functional requirements define the performance, reliability, and usability characteristics of the system.

Performance

The system should process CCTV video streams and generate swimmer counts within 2–5 seconds to ensure near real-time monitoring.

Scalability

The system should support monitoring multiple swimming pools and handle increasing numbers of swimmers without performance degradation.

Security

The system should ensure secure storage and transmission of data through encrypted communication protocols and access control mechanisms.

Reliability

The system should operate continuously with minimal downtime and maintain accurate monitoring even during peak pool usage.

Usability

The pool management dashboard should provide a user-friendly interface that allows operators to easily view swimmer density, water quality parameters, and maintenance alerts.

Data Privacy

The system should avoid storing identifiable personal information and should only process swimmer detection data require for crowd density estimation.

6. COMMERCIALIZATION PLAN

6.1 Target Market and Customer Persona

The proposed crowd-aware pool maintenance scheduling system targets organizations that manage swimming pool facilities and require efficient monitoring and maintenance planning.

The primary target markets include:

- Hotels and resorts with swimming pool facilities
- Public swimming pools operated by local authorities
- Sports complexes and training centers
- Private fitness centers and clubs
- Water parks and recreational facilities

The main users of the system will be pool managers, maintenance staff, and lifeguards, who are responsible for monitoring swimmer activity, maintaining water quality, and ensuring safe swimming conditions.

6.2 Value Proposition

The proposed system provides an intelligent and affordable monitoring solution for swimming pool management.

Key benefits include:

- Real-time swimmer detection using CCTV cameras
- Crowd density estimation for monitoring pool usage
- Smart maintenance scheduling based on real pool conditions
- Reduced chemical waste through optimized dosing
- Improved swimmer safety and hygiene

By replacing traditional fixed maintenance schedules with AI-driven maintenance planning, the system improves operational efficiency and reduces maintenance costs.

6.3 Revenue Model

The system can be commercialized using a low-cost subscription model, making it affordable for small pool facilities.

Possible revenue streams include:

- Monthly subscription for monitoring and maintenance recommendations
- One-time installation fee for system setup
- Optional annual maintenance and technical support services

This model allows pool operators to use the system without large initial investment costs.

6.4 Cost Estimation (Student Prototype – Sri Lanka)

The estimated implementation cost for a basic system prototype is shown below.

Component	Estimated Cost (LKR)
CCTV Camera (Basic IP Camera)	6,000 – 10,000
Water Quality Sensors (Basic set)	12,000 – 18,000
Local Processing Device / Mini PC	20,000 – 30,000
Cloud or Server Setup	5,000 – 8,000
Software Development (Student Prototype)	10,000 – 15,000

Estimate Total Cost : **53,000 – 81,000 LKR**

This low-cost setup makes the system feasible for small pool facilities and student-level prototype implementation.

6.5 Pricing Strategy

To ensure affordability, the system will adopt a tiered subscription model.

Plan	Features	Monthly Price
Basic Plan	Swimmer detection and crowd monitoring	LKR 1,500
Standard Plan	Monitoring + maintenance recommendations	LKR 3,000
Premium Plan	Full analytics, alerts, and dashboard access	LKR 4,500

This pricing strategy ensures that even small swimming pool operators can adopt the system.

6.6 Competitive Advantage

The proposed system offers several advantages compared to traditional pool monitoring methods:

- AI-based swimmer detection using CCTV cameras
- Crowd-aware maintenance scheduling
- Integration of water quality sensor data
- Real-time monitoring dashboard
- Lower operational cost compared to manual monitoring

These features make the system a cost-effective and intelligent solution for modern swimming pool management.

6.7 Intellectual Property Considerations

The proposed system may include intellectual property related to the crowd-aware maintenance scheduling algorithm and AI-based monitoring framework.

Possible protection methods include:

- Copyright protection for the developed software system
- Licensing agreements for commercial deployment
- Potential future patents for intelligent maintenance decision algorithms

Protecting these innovations will help ensure the system can be safely commercialized.

7. BUDGET AND JUSTIFICATION

The development of the proposed crowd-aware pool maintenance scheduling system requires several hardware and software components. The estimated budget for implementing the prototype system is shown below.

7.1 Estimated Project Budget

Item	Description	Estimated Cost (LKR)
CCTV Camera	Basic IP camera for swimmer detection and crowd monitoring	8,000
Water Quality Sensors	Sensors for pH, turbidity, chlorine, and temperature monitoring	15,000
Processing Device	Minicomputer / laptop used for running computer vision models	25,000
Cloud Services	Online storage and server resources for data processing	5,000
Software Tools	Development tools and libraries used for building the system	2,000
Miscellaneous Costs	Networking equipment, cables, and installation materials	5,000

Total Estimated Budget: ~60,000 LKR

7.2 Budget Justification

Hardware Costs

Hardware components such as CCTV cameras and water quality sensors are required to collect real-time data from the swimming pool environment. The CCTV camera captures video streams used for swimmer detection and crowd density estimation, while the sensors monitor important water quality parameters including pH, turbidity, chlorine levels, and temperature.

Software Licenses

Although most development tools used in the project such as Python, OpenCV, and machine learning libraries are open source, a small allocation is included to cover optional software tools or development environments if required.

Cloud Services

Cloud services are required to store collected data, run system analytics, and enable remote monitoring through the pool management dashboard.

Human Effort Cost

The project development is carried out by undergraduate students as part of their academic research. Therefore, the human effort cost is not considered as a commercial development cost but is included as part of the learning and research process.

8. WORK BREAKDOWN STRUCTURE (WBS)

The Work Breakdown Structure (WBS) defines how the tasks of the proposed Water Quality Monitoring and Pool Maintenance System are distributed among the project members. The tasks are organized according to the responsibilities of each member to ensure balanced workload distribution and proper project management.

8.1 Task Breakdown per Member

Member ID	Component		Key Tasks
IT22156310	Crowd-Aware Maintenance Scheduling Module		Analyze swimmer crowd density using CCTV data, estimate crowd density level (CDL), integrate sensor data, generate dynamic maintenance recommendations.
IT22247254	Water Quality Classification and Prediction Module		Collect water quality sensor data, preprocess data, develop machine learning models to classify water quality

			conditions, analyze importance of water quality parameters.
IT22248480	Swimmer Behavior and Drowning Detection Module		Process video streams from CCTV cameras, detect abnormal swimmer behaviors, identify drowning risk situations, generate emergency alerts.
IT22277336	Garbage Intrusion Detection Module		Detect garbage objects entering the pool, analyze human actions related to garbage disposal, trigger alerts for pool management.

8.2 Timeline Alignment

The project tasks are scheduled across different development phases to ensure systematic progress and timely completion of the project.

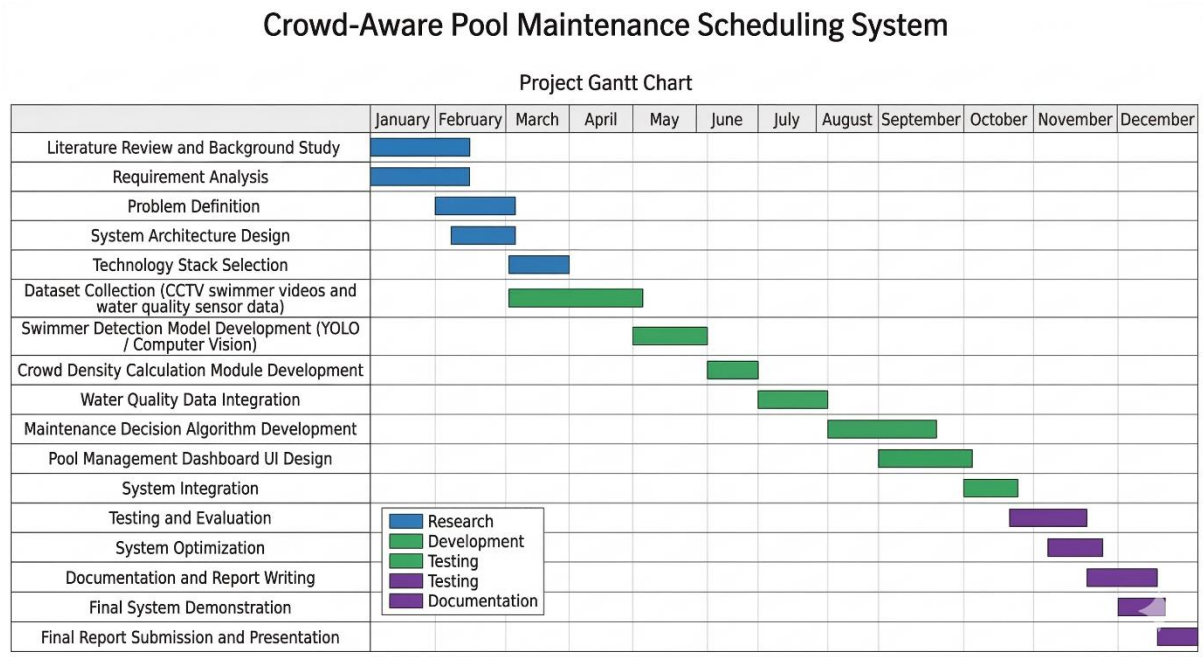
Phase	Activities
Phase 1	Requirement analysis, literature review, and system design
Phase 2	Data collection and dataset preparation
Phase 3	Development of AI models and system modules
Phase 4	Integration of all modules and system testing
Phase 5	Evaluation, documentation, and final project presentation

This phased approach ensures that each module is developed and tested before integration into the complete system.

8.3 Balanced Workload Distribution

The project workload is distributed evenly among the four members, with each member responsible for a specific system component. This structure ensures that all members contribute equally to the overall system development while focusing on their specialized module.
The collaborative integration of these modules results in the complete intelligent pool monitoring and maintenance system.

9 . GANTT CHART



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10. APPENDICES

This section contains supplementary materials that support the proposed Water Quality Monitoring and Pool Maintenance System. These materials provide additional technical details, diagrams, and supporting documents used during the development of the research proposal.

Survey Instruments

This appendix contains the survey questions and data collection instruments used to understand swimming pool management practices, maintenance challenges, and user requirements. The survey helps identify common issues related to water quality monitoring, swimmer crowd density, and maintenance scheduling.

Detailed Diagrams

This appendix includes detailed technical diagrams related to the proposed system, such as the high-level system architecture, CCTV-based swimmer detection workflow, crowd density estimation process, and maintenance scheduling process. These diagrams provide a visual representation of how the system components interact.

Preliminary UI Mockups

This section presents the initial user interface designs of the Pool Management Dashboard. The mockups illustrate how pool managers can monitor swimmer density, water quality parameters, maintenance alerts, and recommended maintenance schedules.

Risk Analysis

This appendix provides an analysis of potential risks associated with the implementation of the proposed system. These risks may include hardware failures, inaccurate swimmer detection, sensor malfunctions, and network connectivity issues. Possible mitigation strategies are also discussed to reduce the impact of these risks.

Ethical Documents

This appendix discusses ethical considerations related to the use of CCTV cameras and data collection in the swimming pool environment. The system is designed to ensure privacy protection by avoiding the storage of personally identifiable information and by using video data only for swimmer detection and crowd density estimation.